

Cloud Security Essentials TASK -3

- 1. Sign up for a free AWS account (or use an existing one), navigate to AWS GuardDuty in the console, and enable GuardDuty for your region.**

Ans :

- Sign in to the AWS Management Console using your AWS account.
 - Select the AWS Region where you want to enable GuardDuty (e.g., ap-south-1 (Mumbai)).
 - In the search bar, type GuardDuty and open the Amazon GuardDuty service.
 - If GuardDuty is not already enabled, click Get Started or Enable GuardDuty.
 - Review the default settings for your account.
 - Click Enable GuardDuty.
 - Wait a few moments while GuardDuty is activated for the selected region.
 - Verify that the GuardDuty Dashboard is displayed and the service status shows Enabled.
- 2. Simulate a suspicious activity by generating a failed login attempt in your AWS account, then check GuardDuty to see if it detects and logs the event.

Hint: Try logging in with the wrong password multiple times or use the AWS CLI to attempt unauthorized actions.**

Ans :

- Sign in to your AWS Management Console.
- Ensure Amazon GuardDuty is enabled.
- Simulate suspicious activity by attempting to log in with an incorrect password several times or perform an unauthorized AWS CLI action using credentials without the required permissions.
- Wait a few minutes for GuardDuty to analyze account activity.
- Open the Amazon GuardDuty console.
- From the left navigation pane, click Findings.
- Check whether any security findings related to the simulated activity are displayed.
- If no findings appear, generate a sample finding:
 1. Open the GuardDuty dashboard.
 2. Click Settings (or Generate sample findings, depending on the console version).
 3. Select Generate sample findings.

3. In Azure, access the Security Center and enable threat protection for a virtual machine or resource group, then review the dashboard for any security alerts or recommendations.

Ans :

- Sign in to the Azure Portal.
- In the search bar, search for Microsoft Defender for Cloud (formerly Azure Security Center) and open it.
- From the left navigation pane, select Environment settings.
- Choose your Azure Subscription.

- Click Defender plans.
- Enable Microsoft Defender for the required resource type (such as Virtual Machines or the relevant workload).
- Click Save to apply the changes.
- Go back to the Microsoft Defender for Cloud dashboard.
- Select Recommendations to view security recommendations for your virtual machine or resource group.
- Open Security alerts to review any detected threats or suspicious activities.
- Review the Secure Score to understand the current security posture of your Azure resources.

4. Compare the types of threats detected by AWS GuardDuty and Azure Security Center by listing 3 examples each, based on your exploration of their dashboards and documentation.

Ans :

AWS GuardDuty	Azure Security Center (Microsoft Defender for Cloud)
Detects unauthorized API calls or suspicious AWS account activity.	Detects malware infections on Azure Virtual Machines.
Detects compromised EC2 instances communicating with malicious IP addresses or domains.	Detects brute-force attacks (such as repeated RDP/SSH login attempts) against virtual machines.

AWS GuardDuty	Azure Security Center (Microsoft Defender for Cloud)
Detects unusual IAM user behavior, credential misuse, or cryptocurrency mining activity.	Detects security misconfigurations, vulnerable resources, and suspicious activities affecting Azure resources.

- Similarities
- Both monitor cloud environments for security threats and suspicious activities.
- Both provide security alerts and recommendations to help users respond to threats.
- Both integrate with their cloud platforms to continuously monitor resources.

5. Describe, step-by-step, how real-time threat detection works in either AWS GuardDuty or Azure Security Center, using a scenario like detecting a brute-force attack on a cloud resource.

Ans :

- Steps
- Enable Amazon GuardDuty in your AWS account for the desired region.
- GuardDuty continuously monitors AWS data sources such as:
 1. AWS CloudTrail logs
 2. VPC Flow Logs
 3. DNS logs

- An attacker starts making multiple failed SSH login attempts on an EC2 instance from the same IP address.
- GuardDuty analyzes the login attempts using machine learning, AWS threat intelligence, and behavior analysis.
- GuardDuty detects that the repeated failed login attempts match the pattern of a brute-force attack.
- GuardDuty generates a security finding with details such as:
 1. Threat type (e.g., SSH Brute Force)
 2. Severity level
 3. Affected EC2 instance
 4. Source IP address
 5. Time of detection
- The finding appears immediately on the GuardDuty Findings Dashboard.
- The security administrator reviews the finding and investigates the affected EC2 instance.
- The administrator takes corrective actions, such as:
 1. Blocking the attacker's IP address using a Security Group or Network ACL.
 2. Changing SSH credentials.
 3. Enabling Multi-Factor Authentication (MFA).
 4. Updating firewall rules and security policies.
- After mitigation, the administrator continues monitoring GuardDuty to ensure no further suspicious activity occurs.

